# AVIRAL NAUTIYAL

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## Education

**M.S. in Computer Science (3.9/4.0)** **Sept. 2024 – May 2026**
*Lakehead University* *Thunder Bay, ON, Canada*
- **Focus areas:** Machine Learning, Deep Learning, NLP, Cloud Computing (AWS), Data Structures & Algorithms
- **Master's Research Project:** Brain Tumor Detection from MRI images using Machine Learning

**Master of Computer Application (8.42/10)** **Aug. 2022 – Jul. 2024**
*DIT University* *Dehradun, Uttarakhand, India*
- **Core Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks
- **Advanced Coursework:** Java Programming, Software Engineering, Artificial Intelligence, Machine Learning

**Bachelor of Science (Physical Science with Computer Science) (8.9/10)** **Aug. 2019 – Jun. 2022**
*University of Delhi* *New Delhi, India*
- **Core Coursework:** Database Management Systems, Operating Systems, Computer System Architecture, R Programming, HTML, Calculus and Matrices, Algebra, Data Structures, Computer Networks, Discrete Mathematics, Probability and Statistics

## Technical Skills

**Programming & Database**: Python (proficient), Java, R, SQL
**Development Tools & Cloud**: Git, GitHub, GitLab, GitHub Actions (CI/CD), Docker, Google Cloud Run, Workload Identity Federation (keyless auth), AWS, Firebase, Gradio, VS Code, Jupyter Notebooks, LaTeX
**Machine Learning & AI**: Deep Learning (CNNs, Transformers), Computer Vision, Transfer Learning, Feature Selection & Engineering, Metaheuristic Optimization, NLP, Model Evaluation, Data Preprocessing
**ML & Data Libraries**: TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
**Backend & APIs**: FastAPI, REST API Design, pytest, Caching
**Platforms**: macOS, Windows, Linux (Ubuntu)
**Methodologies & Soft Skills**: Cross-functional Collaboration, Problem Solving, Communication, Agile/Scrum, Time Management, Adaptability

## Projects

**Brain Tumor Detection using Machine Learning** | | | *Python, TensorFlow, Scikit-learn, Docker, GCP, CI/CD*
- Implemented published **Black Hole Optimization (BHO) feature selection** for brain tumor MRI classification, reducing a **3,840-dimensional fused feature vector** to **2,000 optimal features (∼48% reduction).**
- Fused deep features from **ResNet50, VGG16, and EfficientNet-B0** as the optimization input, improving computational efficiency while preserving classification performance on MRI images.
- Implemented and evaluated an **Optimized Stochastic Gradient Descent based Logistic Regression classifier** across **3 MRI datasets**, achieving up to **98.48% classification accuracy** with significantly **lower testing time** than traditional models. Deployed it as a live, **containerized (using Docker) web demo on Google Cloud Run with GitHub Actions CI/CD pipelines**.

**Canada Labour API** | | | *Python, FastAPI, Docker, Google Cloud Run, GitHub Actions*
- Built a **FastAPI service** over Statistics Canada's Web Data Service with 3 endpoints and 4 labour series (Canada and Ontario, ages 15+ and 15 to 24), returning cleaned JSON with validated inputs and **4 distinct HTTP error codes (404, 422, 502, 503)**.
- Designed **failure handling and caching** for a monthly-updating upstream: retries with exponential backoff and jitter (up to 4 attempts) only for failures that waiting can fix, and a release-aware in-memory cache (24-hour trust, 5-minute recheck after a scheduled release) that serves any 1 to 120 period request from one 120-point fetch per series.
- Wrote **45 offline pytest tests** (recorded replies, fake clock, no network), **containerised it with Docker, and deployed to Google Cloud Run through GitHub Actions using keyless authentication**, with zero stored service account keys.

**Traffic Sign Recognition System** | | *Python, TensorFlow, OpenCV*
- Developed a **CNN-based multi-class image classification system** to recognize **43 traffic sign classes** using the GTSRB dataset.
- Implemented image preprocessing pipelines on **50,000 labelled traffic sign images** including normalization, resizing, and data augmentation techniques.
- Designed and trained deep CNN architectures with regularization techniques and structured training/validation workflows **(early stopping, converging within 20-30 epochs)**, achieving **approximately 97% classification accuracy** on unseen test data.